In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

FIITJEE

JEE (Main), 2017

ALL INDIA TEST SERIES

FULL TEST – IV

Time Allotted: 3 Hours **Maximum Marks: 432**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 03 and 10 to 12)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+8 marks** for correct answer and **– 2 mark** for wrong answer.

Section-A (04 to 09 and 13 to 30) contains 24 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

Straight Objective Type

This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct.

1. The curvilinear motion of a particle is defined by $v_x = 50 - 16t$ and $y = 100 - 4t^2$, where v_x is in meters per second, y is in meters, and t is in seconds. It is also known that x=0 when t = 0. Determine its velocity and acceleration when the position y = 0 is reached.

   (A) $-30\hat{i} - 40\hat{j}$, $-16\hat{i} - 16\hat{j}$ (B) $-40\hat{i} - 30\hat{j}$, $-8\hat{i} - 16\hat{j}$

   (C) $-30\hat{i} - 40\hat{j}$, $-16\hat{i} - 8\hat{j}$ (D) $-40\hat{i} - 30\hat{j}$, $-16\hat{i} - 8\hat{j}$

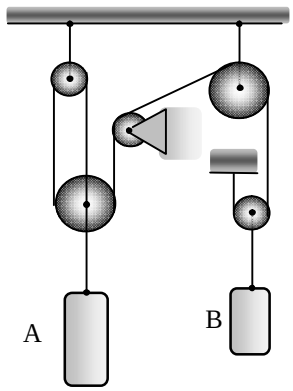
2. A projectile is ejected into an experimental fluid at time t = 0. The initial speed is v_0 and the angle to the horizontal is θ. The drag on the projectile results in an acceleration term a_D = -kv, where k is a constant and v is the velocity of the projectile. What is the terminal velocity? Include the effects of gravitational acceleration.

   (A) $v_0 \cos\theta$ (B) $-\frac{g}{k}$

   (C) $-\frac{v_0}{k}$ (D) Zero

3. At a certain instant, cylinder A has a downward velocity of 0.8 m/s and an upward acceleration of 2 m/s^2. Determine the corresponding velocity and acceleration of cylinder B.

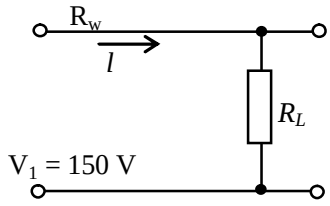
   (A) v_B = 1.2 m/s down, a_B = 3 m/s^2 up
   (B) v_B = 1.2 m/s up, a_B = 3 m/s^2 down
   (C) v_B = 2.4 m/s up, a_B = 6 m/s^2 down
   (D) v_B = 0.6 m/s down, a_B = 1.5 m/s^2 up



Space for rough work

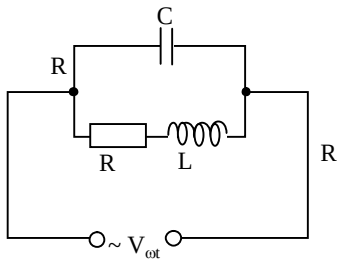
4. The mean value of the kinetic energy and the root-mean-square velocity of helium molecules under normal conditions is close to
(Given: the pressure and temperature of helium under normal conditions are $p_0 = 1.013 \times 10^5 Pa$ and $T_0 = 273K$ respectively. From tables, we find the molar mass of helium, $M = 4 \times 10^{-3}\ kg/mol$, the Boltzmann constant $k = 1.38 \times 10^{-23}\ J/K$, the Avogadro constant $N_A = 6.02 \times 10^{23}\ mol^{-1}$, and the density of helium under normal conditions, $\rho_0 = 0.18\ kg/m^3$.)
(A) 5.65 x 10^{-21} J, 1300 m/s (B) 9.34 x 10^{-21} J,1950 m/s
(C) 5.65 x 10^{-21} J, 1950 m/s (D) 9.34 x 10^{-21} J, 1300 m/s

5. A telegram is sent from Delhi to Allahabad (The distance between the cities of 650 km) via a steel telegraph line carrying a current of 1.7 mA at a voltage of 150 V. The cross-sectional area of the wire is 5.0 mm^2. The voltage drop across the load resistances is (resistivity of steel is 1.2 x 10^{-7} Ω m)



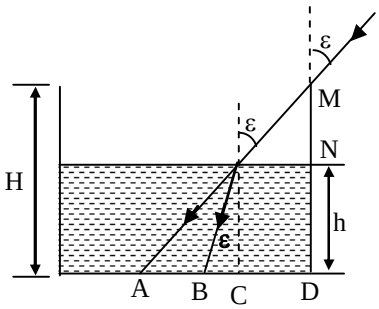
(A) 53 V (B) 97 V
(C) 150 V (D) 0 V

6. Impedance of given circuit is (Given: R = 1 Ω, C = $\frac{5\sqrt{2}}{2\pi} \times 10^{-2}$ Farad, L = $\frac{1}{40\pi}$ Henry, frequency of A.C. source = 100 Hz)



(A) 1 Ω (B) 2 Ω
(C) $\sqrt{2}$ Ω (D) $2\sqrt{2}$ Ω

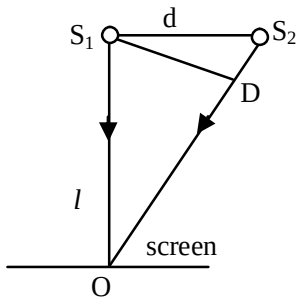
7. The sun forms an angle of 60° with the horizon. Determine the length of the umbra at the bottom of an opaque vessel illuminated by sun light. The height of the vessel is 25 cm. What will the change in the length of the umbra be when water is poured into the vessel to a height of 20 cm shown in figure.
Given: H = 25 cm is the height of the vessel, φ = 60° is the altitude of the Sun above the horizon, h = 20 cm is the height of the water column. From tables, we take the refractive index for water, n = 1.33. The change Δl of the length of the umbra in the vessel with water is



(A) 11 cm (B) 3.4 cm
(C) 14.4 cm (D) 25.4 cm

Space for rough work

8. Two coherent light sources emit light of wavelength 550 nm which produces an interference pattern on a screen (figure).
The sources are 2.2 mm apart and 2.2 m from the screen.
(A) Path difference at point O on screen is 2.2 μm
(B) There is a bright fringe at O
(C) Path difference at point O on screen is 1.1 mm
(D) There is a dark fringe at O



9. A 1 kg object oscillates at the end of a vertical spring that has a spring constant of 100 N/m. The effect of air resistance is represented by the damping coefficient 10 N-s/m. The angular frequency of the damped oscillation is
(A) $10\ s^{-1}$ (B) $5\ s^{-1}$
(C) $5\sqrt{3}\ s^{-1}$ (D) $5\sqrt{2}\ s^{-1}$

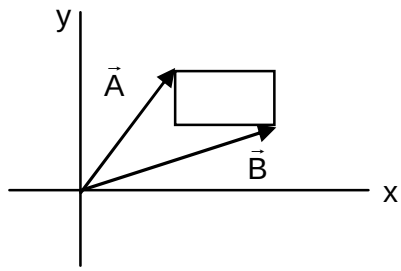
10. A steel wire of length 30.0 m and a copper wire of length 20.0 m, both with 1.00 mm diameters, are connected end to end and stretched to a tension of 150 N. During what time interval will a transverse wave travel the entire length of the two wires?
(A) 0.137 sec (B) 0.192 sec
(C) 0.285 sec (D) 0.329 sec

11. Light of wavelength 587.5 nm illuminates a slit of width 0.75 mm. At what distance from the slit should a screen be placed if the first minimum in the diffraction pattern is to be 0.85 mm from the central maximum?
(A) 2 m (B) 10 m
(C) 1.1 m (D) 2.2 m

12. What is the energy of a photon that, when absorbed by a hydrogen atom, could cause an electronic transition from the n = 2 state to the n = 5 state?
(A) 2.86 eV (B) 0.472 eV
(C) 1.92 eV (D) 10.2 eV

Space for rough work

13. The rectangle shown in figure has sides parallel to the x and y axes. The position vectors of two corners are $\vec{A} = 10.0$ m at 53° and $\vec{B} = 12.0$ m at 30.0^o. Find the perimeter of the rectangle.



(A) 6.2 m (B) 24.8 m
(C) 12.4 m (D) 18.6 m

14. A solid piece of lead has a mass of 23.94 g and a volume of 2.10 cm^3. From these data, calculate the density of lead in SI units (kilograms per cubic meter).
(A) 1.14 x 10^4 kg/m^3 (B) 2.28 x 10^4 kg/m^3
(C) 2.14 x 10^3 kg/m^3 (D) 1.28 x 10^3 kg/m^3

15. A fixed mass of gas is trapped in a syringe. The volume of the syringe is slowly reduced, compressing the gas without changing the temperature. The pressure exerted on walls of syringe charges because
(A) The particles have more kinetic energy
(B) The particles collide with each other more often
(C) The particles hit the walls of the syringe with more force
(D) The particles hit the walls of the syringe more often

16. Light and sound can both be thought of as a wave. Which of the following statements is not true?
(A) They can both transfer energy (B) They can both be polarised
(C) They can both be refracted (D) They can both be diffracted

17. An aircraft of mass 4000 kg produces a thrust of 10 Kn. The aircraft needs to travel at 35 m/s to take off from a standing start, the time to become airborne is approximately.
(A) 2.5 s (B) 3.5 s
(C) 9 s (D) 14 s

Space for rough work

18. The initial activity of a radioactive isotope is 120 Bq and the half-life is 20 minutes. For a sample of the same isotope with twice the mass, the values would be:

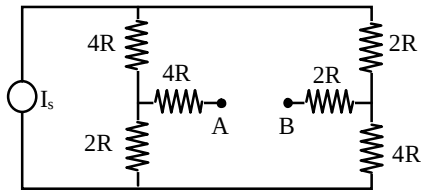
	Initial Activity/Bq	Half life/minutes
(A)	120	20
(B)	240	20
(C)	60	20
(D)	120	40

19. The average wavelength of light emitted from an incandescent torch bulb with a metal filament is 120 nm. Calculate the number of photons emitted by a 20 W torch bulb in one hour.

(A) 2.1×10^{19} (B) 2.1×10^{20}

(C) 12×10^{20} (D) 1.2×10^{19}

20. Consider the circuit shown below. I_s is a constant current source, meaning that no matter what device is connected between points A and B, the current provided by the constant current source is the same.



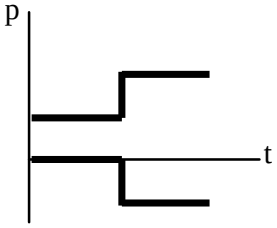
Connect an ideal voltmeter between A and B. Determine the voltage reading in terms of any or all of R and I_s.

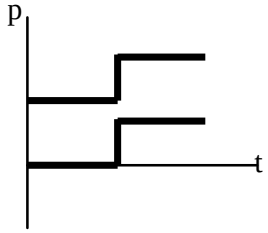
(A) 2iR (B) 0

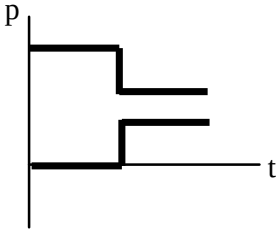
(C) iR (D) $\frac{iR}{2}$

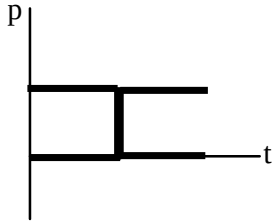
Space for rough work

21. A particle moves along the x-axis. It collides elastically head-on with an identical particle initially at rest. Which of the following graphs could illustrate the momentum of each particle as a function of time?

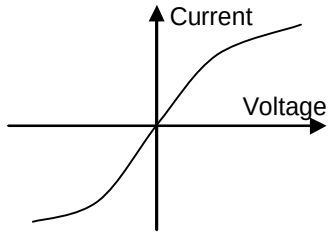
(A) 

(B) 

(C) 

(D) 

22. The voltage – current graph for a filament light bulb is shown below.



The graph has this shape because:
(A) Resistance of the filament increases as the current increases
(B) Resistance of the filament decreases as the current increase
(C) The filament has a higher resistance than the connecting wires
(D) The filament has a lower resistance than the connecting wires

23. A steel ball bearing is dropped onto a concrete floor and rebounds to 80% of its initial height. The percentage of the kinetic energy lost in the collision is
(A) 80% (B) 64%
(C) 36% (D) 20%

Space for rough work

24. An air molecule of mass 5.0×10^{-26} kg and moving with a speed of 500 ms^{-1} has kinetic energy of about $(e = 1.6 \times 10^{-19} C)$

(A) 1 eV (B) 0.4 eV
(C) 40 meV (D) 4 meV

25. The Moon takes 2 minutes to sink below the horizon at the equator when observed at night (about the same time as the Sun takes to set). If the radius of the Earth is 6400 km and the radius of the Moon is 1700 km, what is the angular size of the Earth when observed from the Moon?

(A) 3.8^0 (B) 3.5^0
(C) 1.9^0 (D) 0.3^0

26. A bubble is made from a soap film of uniform thickness enclosing a spherical volume of air. If the diameter of the bubble is trebled, the thickness of the soap film, t, will become

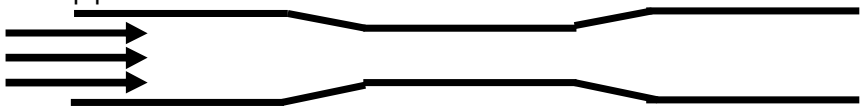
(A) t/27 (B) t/9
(C) t/4 (D) t/3

27. The radius of the Sun is 6.9×10^5 km and that of the Earth is 6400 km. Approximately how many rigid earth spheres could be fitted into the interior of the sun?

(A) 100 (B) 1200
(C) 10000 (D) 10^6

Space for rough work

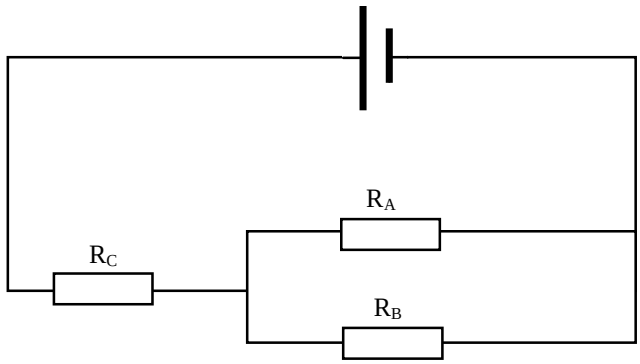
28. An incompressible liquid flows through a pipe which has a section with a narrower bore. What is the speed of the liquid and the volume rate of flow in the narrow section compared to the wider section of the pipe?



	Speed of flow/ms^{-1}	Volume rate of flow /m^3s^{-1}
(A)	Slower	Same
(B)	Faster	Less
(C)	Faster	Same
(D)	Slower	More

29. In the circuit shown below, the power converted in R_A is 1 W. The value of R_A is R, that of R_B is 2R and R_C. How much power is supplied by the cell?

(A) 3W
(B) 4W
(C) 6 W
(D) 8 W



30. The neutral point between the Earth and the Moon is the point where the gravitational pull of the Moon is equal to the gravitational pull of the Earth. If the energy a 1000 kg spacecraft needs in order to reach the neutral point from the Earth is 6.0×10^{10} J and to reach the neutral point from the Moon is 0.25×10^{10}J, what is the minimum energy needed to send a 1 kg rock from the Moon to the Earth?

(A) 0.25×10^7 J
(B) 5.75×10^7 J
(C) 6.0×10^7 J
(D) 6.25×10^7 J

Space for rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct.

1. A mixture contains NaCl and unknown chloride MCl. (a) 1g of this is dissolved in water, excess of acidified $AgNO_3$ solution is added to it, so that 2.567 g of white ppt. is obtained (b) 1 g of original mixture is heated to 300^oC. Some vapours come out which are absorbed in $AgNO_3$ (acidified) solution. 1.341 g of white precipitate is obtained. Find molecular weight of unknown chloride.
   (A) 53.50 (B) 45.5
   (C) 58.5 (D) none of these

2. At 20^oC, two balloons of equal volume and porosity are filled to a pressure of 2 atm, one with 14 kg N_2 and other with 1 kg of H_2. The N_2 balloon leaks to a pressure of ½ atm in 1 hr. How long will take for H_2 balloon to reach a pressure of ½ atm?
   (A) 4 min (B) 12 min
   (C) 16 min (D) 20 min

3. When photons of energy 4.25 eV strike the surface of a metal A, the ejected photoelectrons have maximum kinetic energy, T_A (expressed in eV) and de Broglie wavelength λ_A. The maximum kinetic energy of photoelectrons liberated from another metal B by photons of energy 4.70 eV is $T_B = T_A - 1.50$ eV. If the de Broglie wavelength of these photoelectrons is $\lambda_B = 2\lambda_A$, then which is not correct
   (A) The work function of A is 2.25 eV (B) the work function of B is 1.20 eV
   (C) $T_A = 2.00$ eV (D) $T_B = 0.5$ eV

Space for rough work

4. A radioactive isotope decays as ${}^{m}_{z}A \xrightarrow{} {}^{m-4}_{z-2}B \xrightarrow{} {}^{m-4}_{z-1}C$. The half-lives of A and B are 10 month and 6 month respectively. Assuming that initially only A was present, will it be possible to achieve radioactive equilibrium for B. If so what would be the ratio of A and B at equilibrium? What would happen if the half-lives A and B were 6 month and 10 month respectively?

(A) 1.66, no equilibrium

(B) 1.77, $t_{\frac{1}{2}A} > t_{\frac{1}{2}B}$

(C) 2.77, $\lambda_B > \lambda_A$

(D) 2.88 $\lambda_B = \lambda_A$

5. An electric current is passed through two electrolytic cells connected in series, one containing $AgNO_3$ (aq) and other H_2SO_4 (aq). What volume of O_2 measured at 25°C and 750 mm Hg would be liberated from H_2SO_4, if

(a) 1 mole of Ag^+ are deposited from $AgNO_3$ solution

(b) 8×10^{22} ions of Ag^+ are deposited from $AgNO_3$ solution

(A) 6.2, 0.823

(B) 3.27, 0.78

(C) 3.1, 5.26

(D) 1.12, 3.17

6. 1 g of Mg atoms in the vapour phase absorbs 50.0 kJ of energy. Find the composition of Mg^+ and Mg^{2+} formed as a result of absorption of energy. IE_1 and IE_2 for Mg are 740 and 1450 kJ mol^{-1} respectively.

(A) 68.28%, & 31.72%

(B) 31.72% & 68.28%

(C) 70% & 30%

(D) 20% & 80%

7. The following compounds have been arranged in order of their increasing thermal stabilities. Identify the correct order:

K_2CO_3 (I) $MgCO_3$(II) $CaCO_3$(III) $BeCO_3$(IV)

(A) I < II < III < IV

(B) IV < II < III < I

(C) IV < II < I < III

(D) II < IV < III < I

Space for rough work

8. The freezing point of a solution containing 0.2 g of acetic acid in 20.0 g benzene is lowered by 0.45°C. Calculate the degree of association of acetic acid in benzene. Assume acetic acid dimerizes in benzene. K_f for benzene 5.12 K mol^{-1} kg.
(A) 46.6% (B) 94.6%
(C) 86.6% (D) 37.6%

9. For the system $A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$, $\Delta H = -80$ KJ/mole
If the activation energy for the forward step is 100 kcal/mole. What is the ratio of temperature at which the forward and backward reaction shows the same % change of rate constant per degree rise of temperature ? (1 cal = 4.2 J)
(A) $\sqrt{0.72}$ (B) $\sqrt{0.84}$
(C) $\sqrt{0.42}$ (D) $\sqrt{1}$

10. At 273 K and one atm, 'a' litre of N_2O_4 decomposes to NO_2 as : $N_2O_4(g) \rightleftharpoons 2NO_2(g)$. To what extent has the decomposition proceeded when the original volume is 25% less than that of existing volume?
(A) 25% (B) 30%
(C) 60% (D) 33%

11. N_2O_4 dissociates as : $N_2O_4 \rightleftharpoons 2NO_2$. At 55°C and one atmosphere, N_2O_4 was decomposed to 50.3%. At what P and temperature, the equilibrium mixture has the ratio of N_2O_4: NO_2 as 1 : 8?
(A) 2.18 (B) 0.19
(C) 3.7 (D) 1.18

12. An equilibrium mixture at 300 K contains N_2O_4 and NO_2 at 0.28 and 1.1 atmosphere respectively. If the volume of container is doubled, calculate the new equilibrium pressure of two gases.
(A) 4.32 (B) 4.68
(C) 2.4 (D) 1.98

Space for rough work

13. Calculate the amount of NH_3 and NH_4Cl required to prepare a buffer solution of pH 9.0 when total concentration of buffering reagents is 0.6 mol L^{-1}.
$pK_b \ for \ NH_3 = 4.7, \log 2 = 0.30$
(A) 0.2 M & 0.4 M
(B) 0.4 M & 0.4 M
(C) 0.2 & 0.2 M
(D) 2 M & 4 M

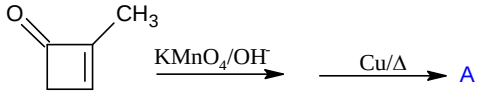
14. For a reaction,
$$M_2O(s) \longrightarrow 2M(s) + \frac{1}{2}O_2(g)$$
ΔH=30 kJ mol^{-1} and ΔS=0.07 kJ K^{-1} mol^{-1} at 1 atm. Calculate upto which temperature, the reaction would not be spontaneous.
(A) 528 K
(B) 428 K
(C) 480 K
(D) 820 K

15. Dipole moment will be zero in complexes
$I : \left[Ni(CN)_4\right]^{2-} \qquad II : cis - Pt\left[(NH_3)_2 Cl_2\right] \qquad III : trans - \left[Pt(NH_3)_2 Cl_2\right]$
(A) I and II
(B) I and III
(C) II and III
(D) I, II and III

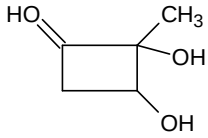
16. Primary alkyl halide C_4H_9Br (A) reacted with alcoholic KOH to give compound (B). Compound (B) is reacted with HBr to give (C) which is an isomer of (A). When (A) is reacted with sodium metal it gives compound (D), C_8H_{18} which is different from the compound formed when n-butyl bromide is reacted with sodium. Give the structural formula of (A).
(A) n-butyl bromide
(B) isobutyl bromide
(C) 1-bromo 2-methyl propane
(D) butane

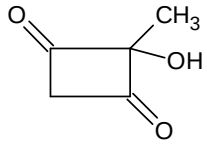
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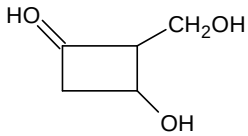
17. Consider the following reaction,

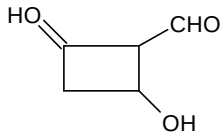


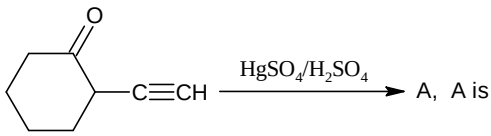
A formed is

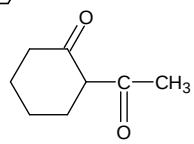
(A) 

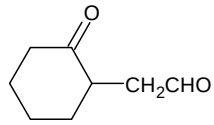
(B) 

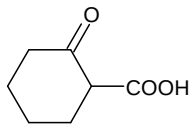
(C) 

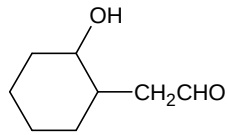
(D) 

18. 

(A) 

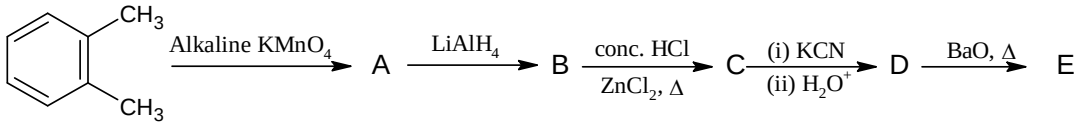
(B) 

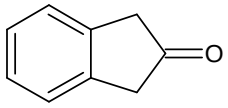
(C) 

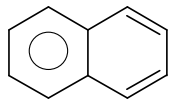
(D) 

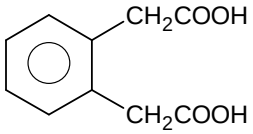
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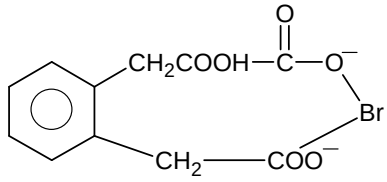
19. Identify E in the following:



(A) 

(B) 

(C) 

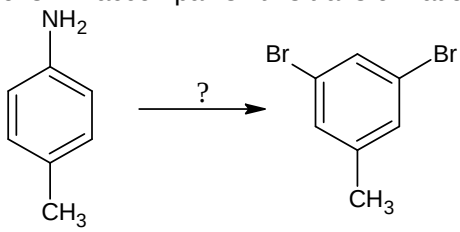
(D) 

20. Amongst the following, total number of essential amino acids are:
valine, glycine, phenylalanine, lysine, leucine, isoleucine, methionine, alanine, threonine.

(A) 3 (B) 7

(C) 5 (D) 9

21. Which sequence of reactions will accompanish this transformation?

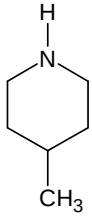


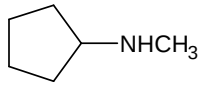
(A) $NaNO_2/HCl/0^oC$, H_3PO_2 and Br_2 (B) Br_2, $LiAlH_4$ and H_3O^+

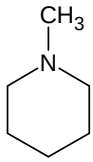
(C) $NaNO_2/HCl/0^oC$, CuBr (D) Br_2, $NaNO_2/HCl/0^oC$, H_3PO_2

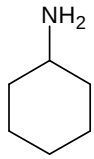
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22. The amines shown are isomers. Choose the one with the lowest boiling point.

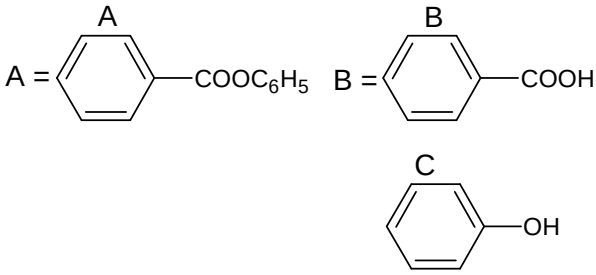
(A) 

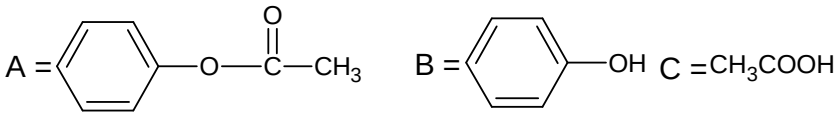
(B) 

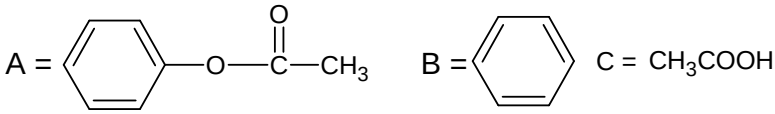
(C) 

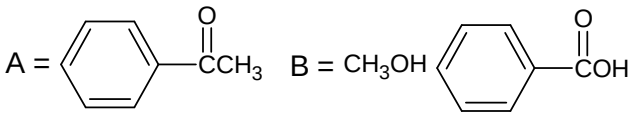
(D) 

23. An ester (A) on acidic hydrolysis gives (B) and (C). (B) gives violet colour and (C) gives blood red colour with neutral $FeCl_3$ solution. (A), (B) and (C) are

(A) 

(B) 

(C) 

(D) 

Space for rough work

24. An organic compound with the molecular formula $C_9H_{10}O$ forms 2, 4-DNP derivative, reduces Tollen's reagent and undergoes Cannizzaro reaction. On vigorous oxidation, it give 1, 2-benzenedicarboxylic acid. Identify the compound.

(A) 2 ethyl benzaldehyde
(B) 3-ethyl benzoic acid
(C) 2 methyl benzaldehyde
(D) 3-ethyl benzaldehyde

25. From the following compounds, how many compounds are soluble in water?

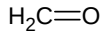
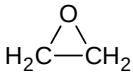
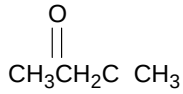
$C_6H_5OH\left(NO_2\right)$, $C_6H_5COC_6H_5$, C_6H_5Cl, C_6H_5CHO, C_6H_5OH.

(A) 2
(B) 3
(C) 1
(D) 4

26. An organic compound (X) has molecular formula $C_7H_6O_2$ and it does not effervesces with $NaHCO_3$. A on treatment with excess of HBr (aq) gives Y($C_6H_6O_2$) and CH_2Br_2. Y forms a violet coloured solution with $FeCl_3$. If X is treated with Br_2/Fe, How many different monobromination product would result?

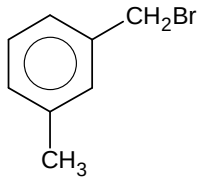
(A) 4
(B) 3
(C) 2
(D) 4

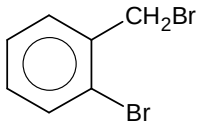
27. An organic compound B is formed by the reaction of ethyl magnesium iodide (CH_3CH_2MgI) with a substance A, followed by treatment with dilute aqueous acid. Compound B does not react with PCC or PDC in dichloromethane. Which of the following is a possible candidate for A?

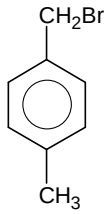
(A) 
(B) 
(C) 
(D) 

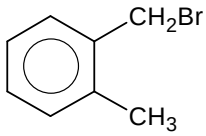
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28. Compound (A), C_8H_9Br gives a white precipitate when warmed with alcoholic $AgNO_3$. Oxidation of (A) gives and acid (B), $C_8H_6O_4$. (B) easily forms anhydride on heating. Identify the compound (A).

(A) 

(B) 

(C) 

(D) 

29. In XeF_4, no. of lone pair e^- are

(A) 2 (B) 3

(C) 5 (D) 4

30. 1 mole each of H_3PO_2, H_3PO_3 and H_3PO_4 will neutralise x moles of NaOH, y moles of $Ca(OH)_2$ and z moles $Al(OH)_3$ (assuming all as strong electrolytes). x, y, z are in the ratio of

(A) 3 : 15 : 1 (B) 1 : 2 : 3

(C) 3 : 2 : 1 (D) 1 : 1 : 1

Space for rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct.

1. The total number of local maxima and local minima of f(x) = x $|x^2 - x| + |x - 2|$ is
   (A) 2 (B) 3
   (C) 4 (D) 0

2. If $f(x) = x^2 + \frac{1}{x}\int_1^x f(t)\,dt$ then total number of points where $|f(x)|$ is not differentiable.
   (A) 0 (B) 1
   (C) 2 (D) 3

3. If f(x) is a differentiable function for all $x \geq 1$, f(1) = 1 and $f'(x) > 2f(x)$, the total number of solution of the equation f(x) = e^{2x-2} in $x \in (1, \infty)$ is,
   (A) 1 (B) 2
   (C) 0 (D) 4

4. If $A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ and $A(adj(A)) = k\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ the K is
   (A) 1 (B) 3
   (C) 5 (D) 4

5. Number of real roots of the equation $f(x) = x^2e^x - 2e^x + 3$ in (-4, 0) is
   (A) 0 (B) 1
   (C) 2 (D) 3

Space for rough work

6. The number of solution of the equation $e^x = 3x^2$ is
(A) 0 (B) 1
(C) 2 (D) 3

7. Let tangent drawn form the point (3, 2) to the circle $x^2 + y^2 - 2x - 2y = 0$ meet at the points A and B. The ortho centre of triangle OAB is where 'O' is origin.
(A) $\left(\frac{67}{25}, \frac{56}{25}\right)$ (B) $\left(\frac{67}{25}, \frac{-1}{25}\right)$
(C) $\left(\frac{57}{25}, \frac{-4}{25}\right)$ (D) $\left(\frac{57}{25}, \frac{4}{25}\right)$

8. The value of $\frac{\int_0^1 \left(1-x^n\right)^m}{\int_0^1 \left(1-x^n\right)^{m+1}}$ is (where m,n $\in I^t$)
(A) $\frac{m+1}{mn+1}$ (B) $\frac{mn+n+1}{mn+n}$
(C) $\frac{mn+n-1}{mn+n}$ (D) $\frac{n+1}{mn+n}$

9. The point on the curve which is nearest to line y = 2x – 1
(A) (1, 2) (B) (1,3)
(C) (-1, 3) (D) (-1, 2)

10. The number of complex numbers which satisfy the equation $z^2 = iz + |z| - 2$
(A) 4 (B) 2
(C) 3 (D) 1

Space for rough work

11. The general solution of the differential equation $ydx - xdy + 3x^2y^2\ e^{x^3}\ dx = 0$, is

(A) $\frac{x}{y} - e^{x^3} = C$ (B) $\frac{x}{y} + e^{x^3} = C$

(C) $\frac{y}{x} - e^{x^3} = C$ (D) $\frac{y}{x} + e^{x^3} = C$

12. If $f(a) = 0, f(b) = 3, f(c) = -1, f(d) = 2, f(e) = 0$ and

h(x) = $x\left(f'(x)\right)^2 + \left(x^2 - 1\right) f'(x)\, f''(x)$ then min number of zeros of h(x) is,

(A) 8 (B) 5
(C) 6 (D) 10

13. f(x) is polynomial of 8th degree satisfies f(x) = f(1- x) $\forall n \in R$ if f(x) = 0 has 6 distinct roots and two equal roots then sum of the roots are

(A) 4 (B) 8
(C) 12 (D) 2

14. If $f(x) = x^5 + 2x - 4$

If $g(x) = f^{-1}(x)$

Then value of $f'\left(g'(x)\right)$. at x = - 4 is

(A) $\frac{35}{16}$ (B) $\frac{37}{16}$

(C) $\frac{1}{2}$ (D) $\frac{-4}{16}$

15. The equation of circle having least radius which touches the line y = 2x at the point (1, 2) and also touches the x axis is,

(A) $x^2 + y^2 - 2\sqrt{5}x - y\left(5 - \sqrt{5}\right) + 5 = 0$ (B) $x^2 + y^2 + 2\sqrt{5}x - y\left(5 - \sqrt{5}\right) + 5 = 0$

(C) $x^2 + y^2 + 2\sqrt{5}x + y\left(5 - \sqrt{5}\right) + 5 = 0$ (D) $x^2 + y^2 + 2\sqrt{5}x + y\left(5 + \sqrt{5}\right) + 5 = 0$

Space for rough work

16. Resolved part of vector $\vec{a}$ along the vector $\vec{b}$ is $\vec{a_1}$ and that perpendicular to $\vec{b}$ is $\vec{a}_2$ then $\vec{a_1} \times \vec{a_2}$ is equal to

(A) $\dfrac{\left(\vec{a} \times \vec{b}\right)\vec{b}}{\left|\vec{b}\right|^2}$ (B) $\dfrac{\left(\vec{a} \cdot \vec{b}\right)\vec{a}}{\left|\vec{a}\right|^2}$

(C) $\dfrac{\left(\vec{a} \cdot \vec{b}\right)\left(\vec{b} \times \vec{a}\right)}{\left|\vec{a}\right|^2}$ (D) $\dfrac{\left(\vec{a} \cdot \vec{b}\right)\left(\vec{b} \times \vec{a}\right)}{\left|\vec{b}\right|^2}$

17. Angle between vector $\vec{a}$ and $\vec{b}$ where $\vec{a}, \vec{b}, \vec{c}$ are unit vectors satisfying $\vec{a} + \vec{b} + \sqrt{3}\vec{c} = 0$ is

(A) $\pi/3$ (B) $\pi/6$

(C) $2\pi/3$ (D) $\pi/12$

18. Volume of parallelepiped determined by vectors $\vec{a}, \vec{b}$ and $\vec{c}$ is 4 unit, Then the volume of the parallelepiped determined by vectors. $\vec{a} \times \vec{b}, \vec{c} \times \vec{a}, \vec{b} \times \vec{c}$ is,

(A) 8 (B) 16

(C) 32 (D) 64

19. If f(x) = $\left(1+x\right)\left(1+\int_1^x \dfrac{f^2(t)}{1+t}dt\right)$, then f(7) is

(A) -1 (B) -5

(C) 1 (D) 4

Space for rough work

20. The value of $\sum_{r=1}^{n}(-1)^{r-1}\left(\frac{r}{r+1}\right)\cdot {}^{n}C_{r}$ is

(A) $\frac{1}{n+1}$ (B) $\frac{1}{n+2}$

(C) $\frac{1}{n-1}$ (D) $\frac{n+1}{n+2}$

21. If $a_1, a_2, a_3 \ldots\ldots a_{11}$ are in A.P if $a_6 = 10$ then value of $\sum_{i=1}^{11} a_i$ is

(A) 110 (B) 50

(C) 100 (D) 95

22. If $\left(1+x+x^2\right)^n = a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots. + a_{2n}x^{2n}$ Then $a_0 + a_3 + a_6 + \ldots..is$

(A) 3^{n-1} (B) 3^n

(C) 2^n (D) 2

23. If z_1 and z_2 are two complex numbers such that $z_1, z_2 \neq 0$, If $\left|\frac{z_1 - z_2}{z_1 + z_2}\right| = 1$, then $\left|Arg\left(\frac{z_1}{z_2}\right)\right|$ is

(A) $\frac{\pi}{3}$ (B) π

(C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

24. If |z| = 2 the locus of $w = z + \frac{1}{z}$ is,

(A) ellipse with eccentricity = 2/5 (B) ellipse with e = 4/5

(C)Hyperbola with e = 5/4 (D) Pe (z) = 0

Space for rough work

25. $\sum_{n=2}^{49} \frac{1}{2^{n-1}+2^{-n}-1.5}$ is

(A) $4-\frac{2^{50}}{2^{49}-1}$ (B) $6-\frac{2^{50}}{2^{49}-1}$

(C) $8-\frac{2^{50}}{2^{49}-1}$ (D) None of these

26. The value of integral $\int_0^{100+m} \{2x\} e^{\{2x\}} dx$ is

Where $O < m < 1/2$ is

(A) $\frac{e^{2m}(2m-1)+201}{2}$ (B) $\frac{e^{2m}(m-1)+201}{2}$

(C) $\frac{e^{2m}+201}{2}$ (D) $\frac{(m+1)e^{2m}+201}{2}$

27. The equation of tangents drawn from the point (0,4) to the hyperbola $x^2-4y^2=36$ are

(A) $5x-6y+24=0$ and $5x+6y-24=0$ (B) $x-4y+16=0$ and $x+4y-16=0$

(C) $2x-y+4=0$ and $2x+y-4=0$ (D) none of these

Space for rough work

28. There are exactly two points on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ whose distance from the center of the ellipse are equal and equal to $\sqrt{\frac{a^2+2b^2}{2}}$. Eccentricity of this ellipse is equal to

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{\sqrt{3}}$

(C) $\frac{1}{\sqrt{2}}$ (D) $\sqrt{\frac{2}{3}}$

29. The sum of two natural numbers m and n is equal to 100, The probability that their product being greater than 1600 is,

(A) $\frac{10}{94}$ (B) $\frac{13}{99}$

(C) $\frac{28}{99}$ (D) $\frac{59}{99}$

30. If x_1 is a solution $2^{x^2+2x+3} \le 4-\left(x^2-1\right)^2$ then . $\lim_{x\to x_1}\left(1+\sin \pi x\right)^{\tan\frac{x\pi}{2}}$

(A) e^{-2} (B) e

(C) e^{-1} (D) e^2

Space for rough work